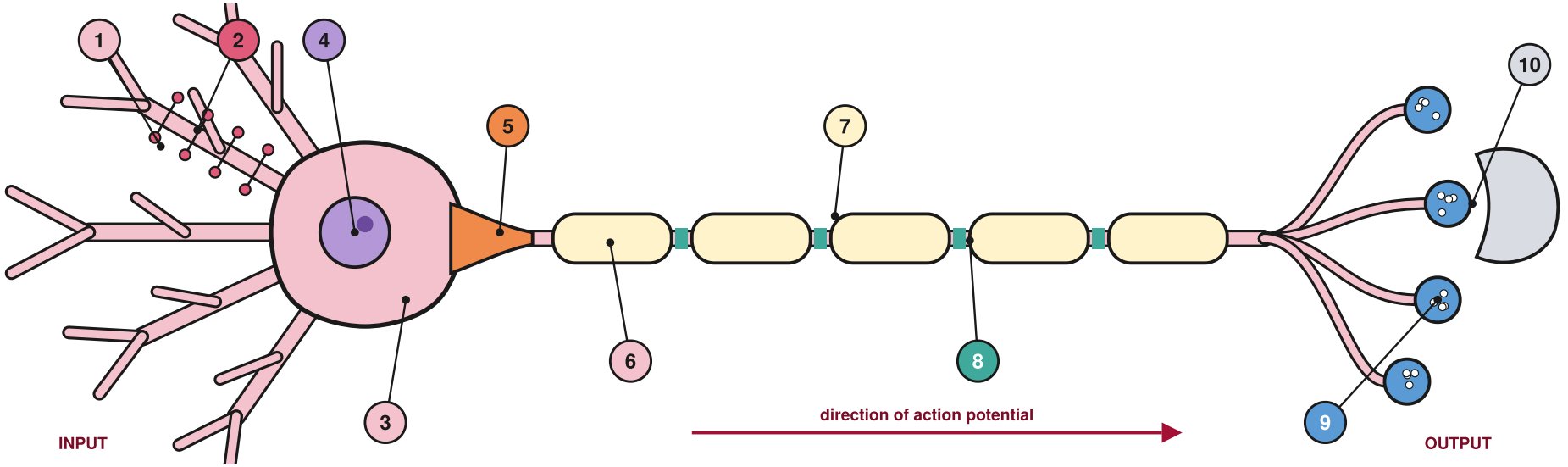
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Neuron Anatomy — Parts & Roles

MCB80x · input → integration → conduction → output

DIAGRAM — MYELINATED NEURON



Schematic, not to scale. Signal flows left to right: dendrites → soma → axon → terminals.

#	Structure	Function · exam fact
1	**Dendrites** Input zone	Branches that receive signals from other neurons. **FACT** Graded EPSPs/IPSPs spread passively and shrink with distance.
2	**Dendritic spines** Input zone	Tiny protrusions holding most excitatory synapses. **FACT** Spines grow and reshape with learning (LTP).
3	**Soma (cell body)** Integration	Holds the nucleus and protein-making machinery; sums incoming inputs. **FACT** Spatial + temporal summation decide whether to fire.
4	**Nucleus** Integration	Stores DNA; controls gene expression and protein synthesis. **FACT** Long-term memory needs new gene expression.
5	**Axon hillock** Trigger zone	Packed with voltage-gated Na^+ channels; the action potential starts here. **FACT** Fires when membrane reaches threshold (~−55 mV).

#	Structure	Function · exam fact
6	**Axon** Conduction	Carries the action potential away from the soma. **FACT** Wider axons conduct faster; some reach ~1 m long.
7	**Myelin sheath** Conduction	Fatty glial wrapping: oligodendrocytes (CNS), Schwann cells (PNS). **FACT** Multiple sclerosis destroys CNS myelin, slowing conduction.
8	**Node of Ranvier** Conduction	Unmyelinated gaps dense with Na^+ channels where the signal regenerates. **FACT** Saltatory conduction: AP "jumps" node to node.
9	**Axon terminals** Output	Boutons full of vesicles that release neurotransmitter. **FACT** Ca^{2+} entry triggers vesicle fusion and release.
10	**Synapse** Output	Junction: presynaptic terminal, cleft, postsynaptic membrane. **FACT** Cleft is ~20 nm wide; transmitter diffuses across.

Personal MCB80x study chart — not an official HarvardX/edX publication.